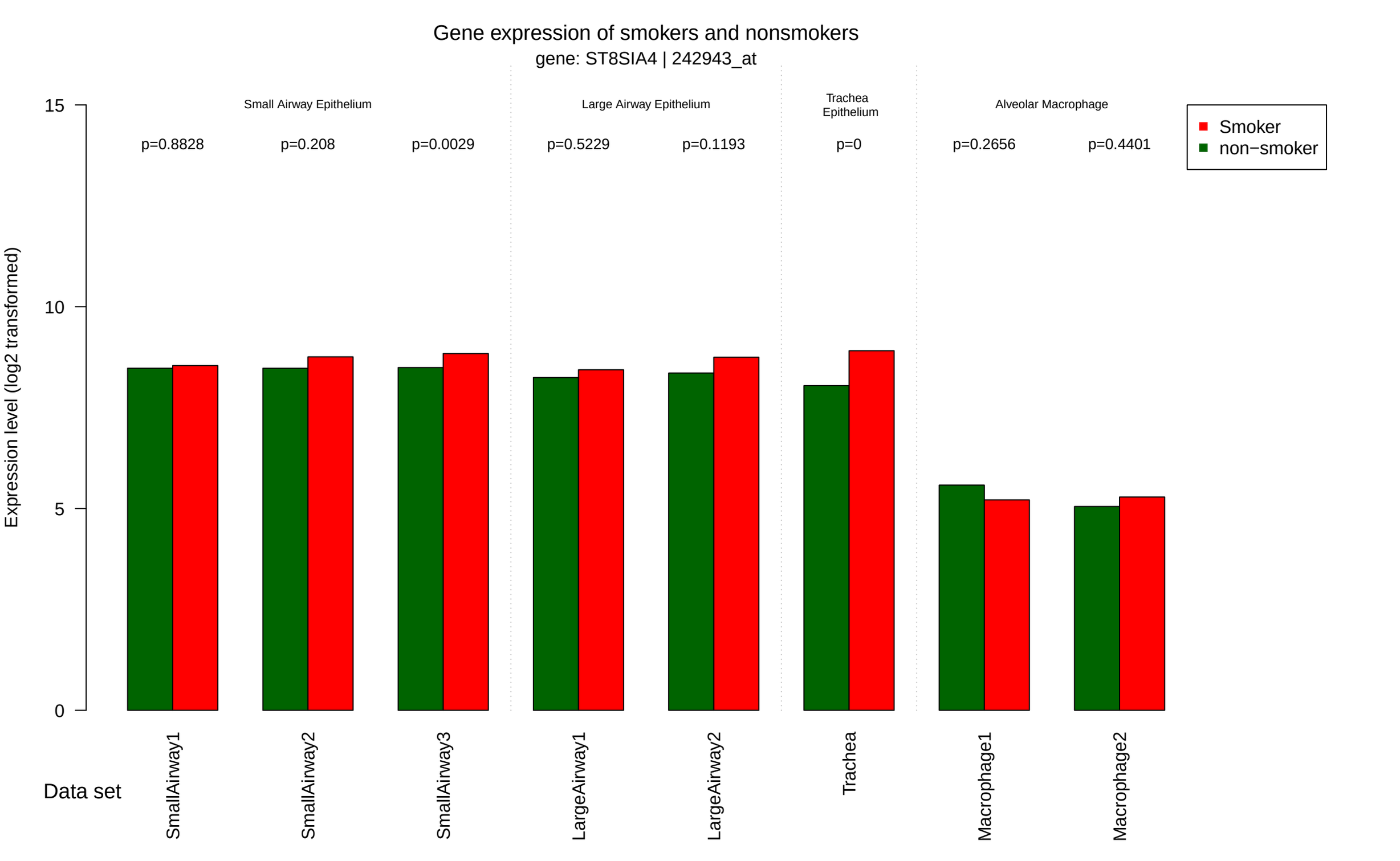
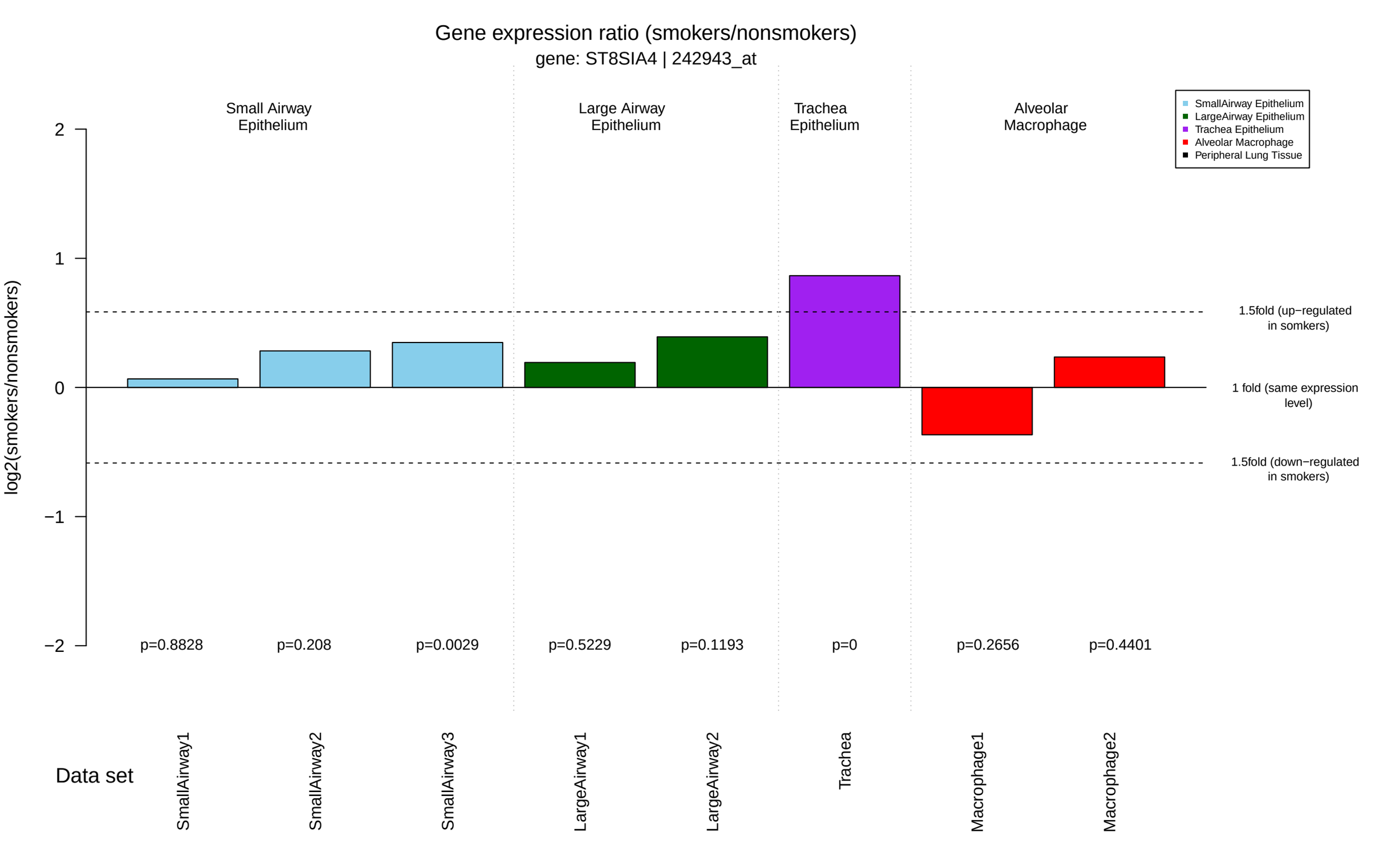
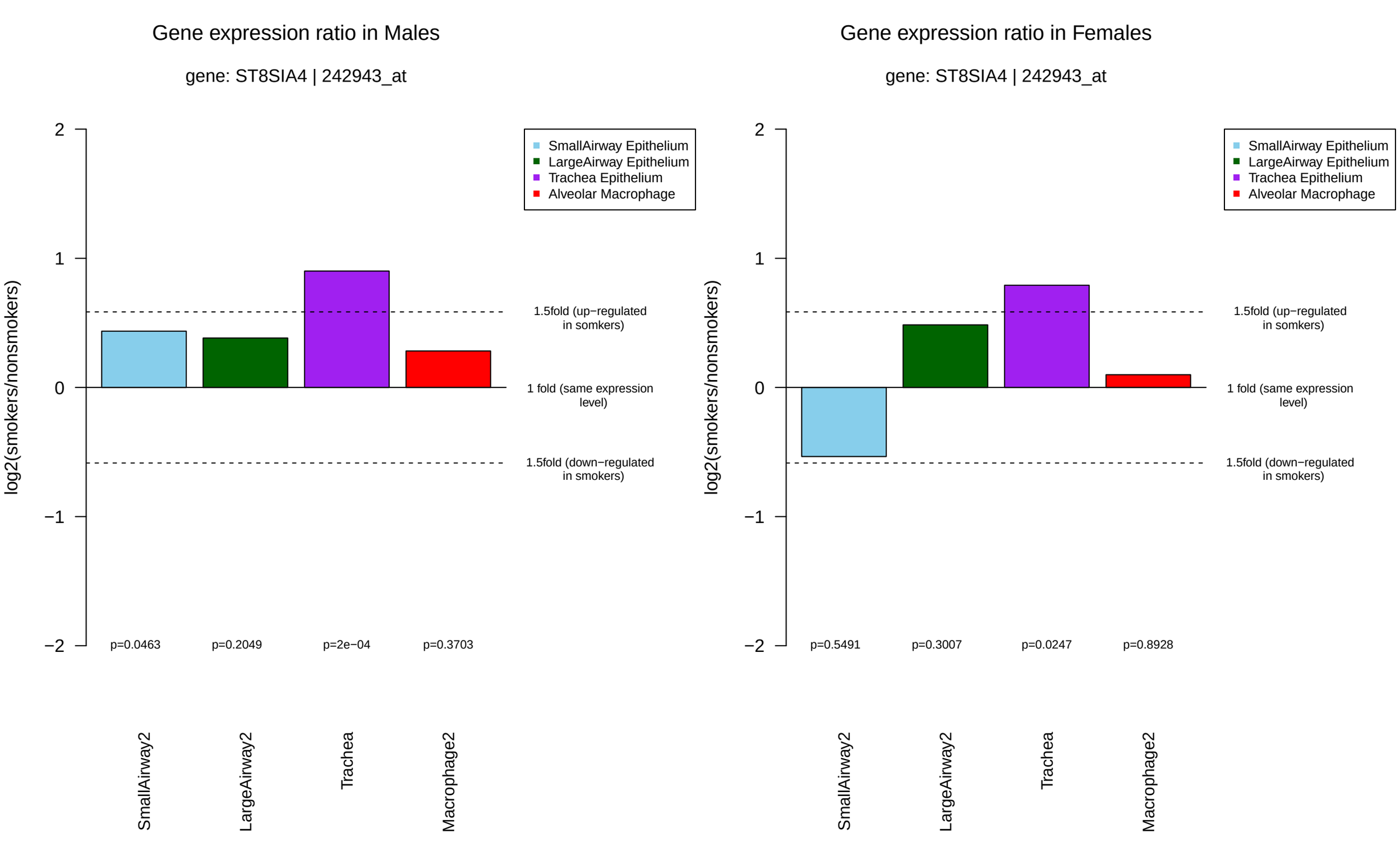
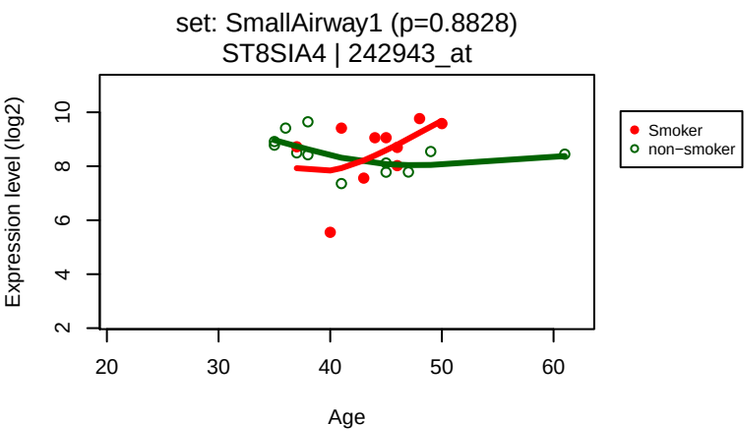
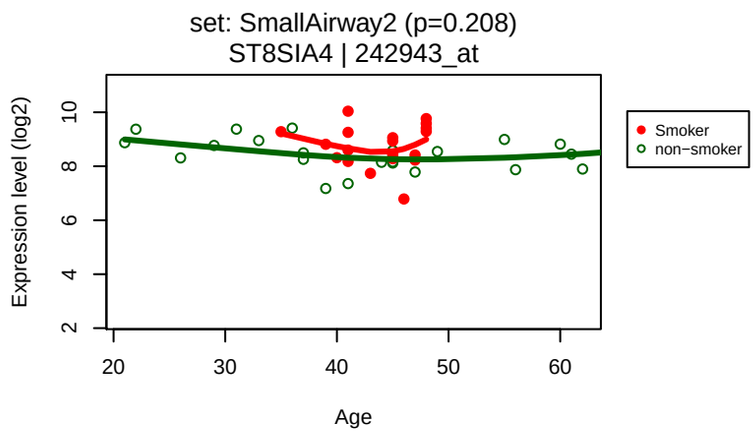
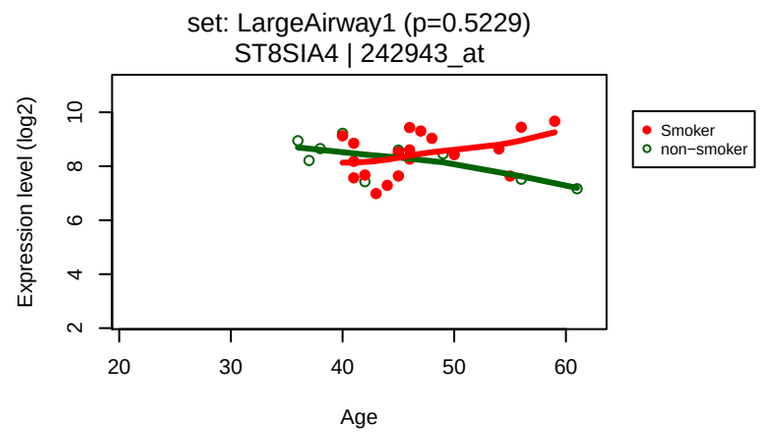

Gene expression of smokers and nonsmokers
gene: ST8SIA4 | 242943_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8828
p=0.208
p=0.0029
p=0.5229
p=0.1193
p=0
p=0.2656
p=0.4401
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

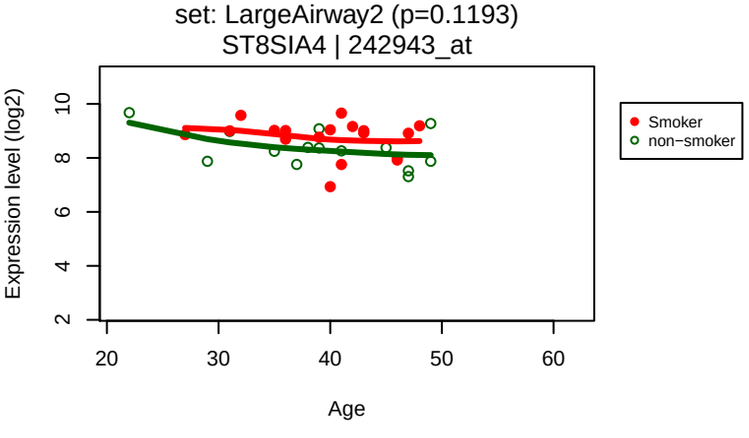

Gene expression ratio (smokers/nonsmokers)
gene: ST8SIA4 | 242943_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8828
p=0.208
p=0.0029
p=0.5229
p=0.1193
p=0
p=0.2656
p=0.4401
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

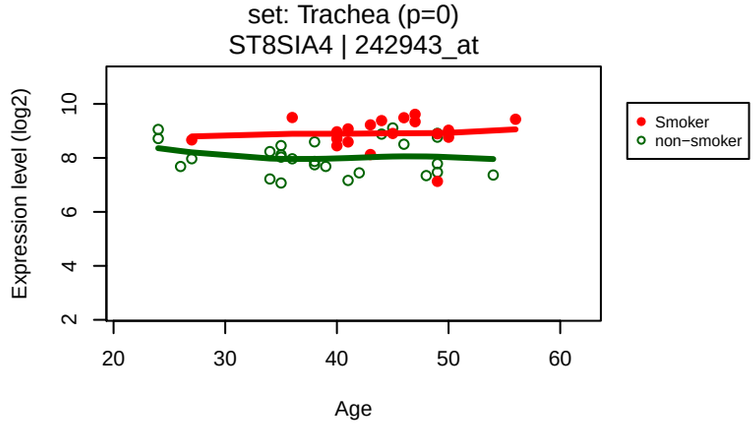

Gene expression ratio in Males
gene: ST8SIA4 | 242943_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0463
p=0.2049
p=2e−04
p=0.3703
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST8SIA4 | 242943_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5491
p=0.3007
p=0.0247
p=0.8928
SmallAirway2
LargeAirway2
Trachea
Macrophage2

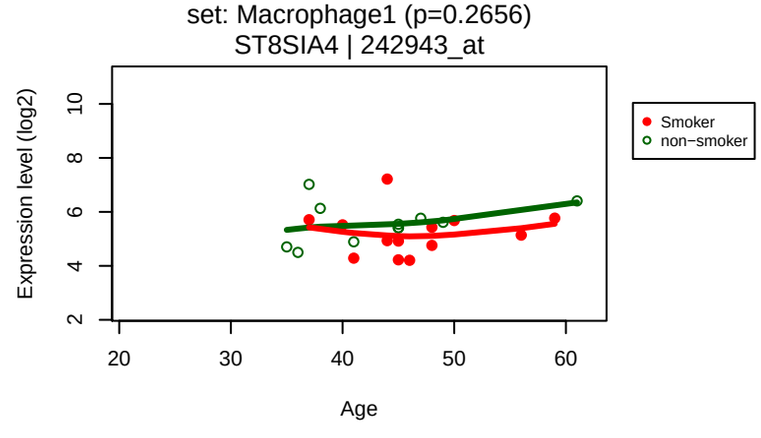

set: SmallAirway1 (p=0.8828)
ST8SIA4 | 242943_at
Expression level (log2)
Age
Smoker
non−smoker

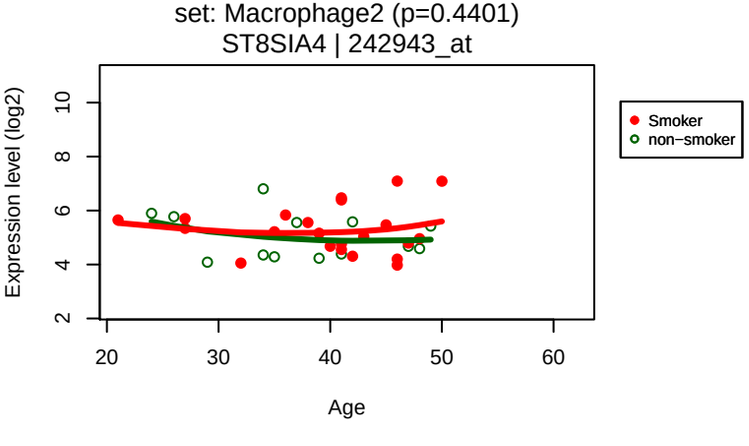

set: SmallAirway2 (p=0.208)
ST8SIA4 | 242943_at
Expression level (log2)
Age
Smoker
non−smoker

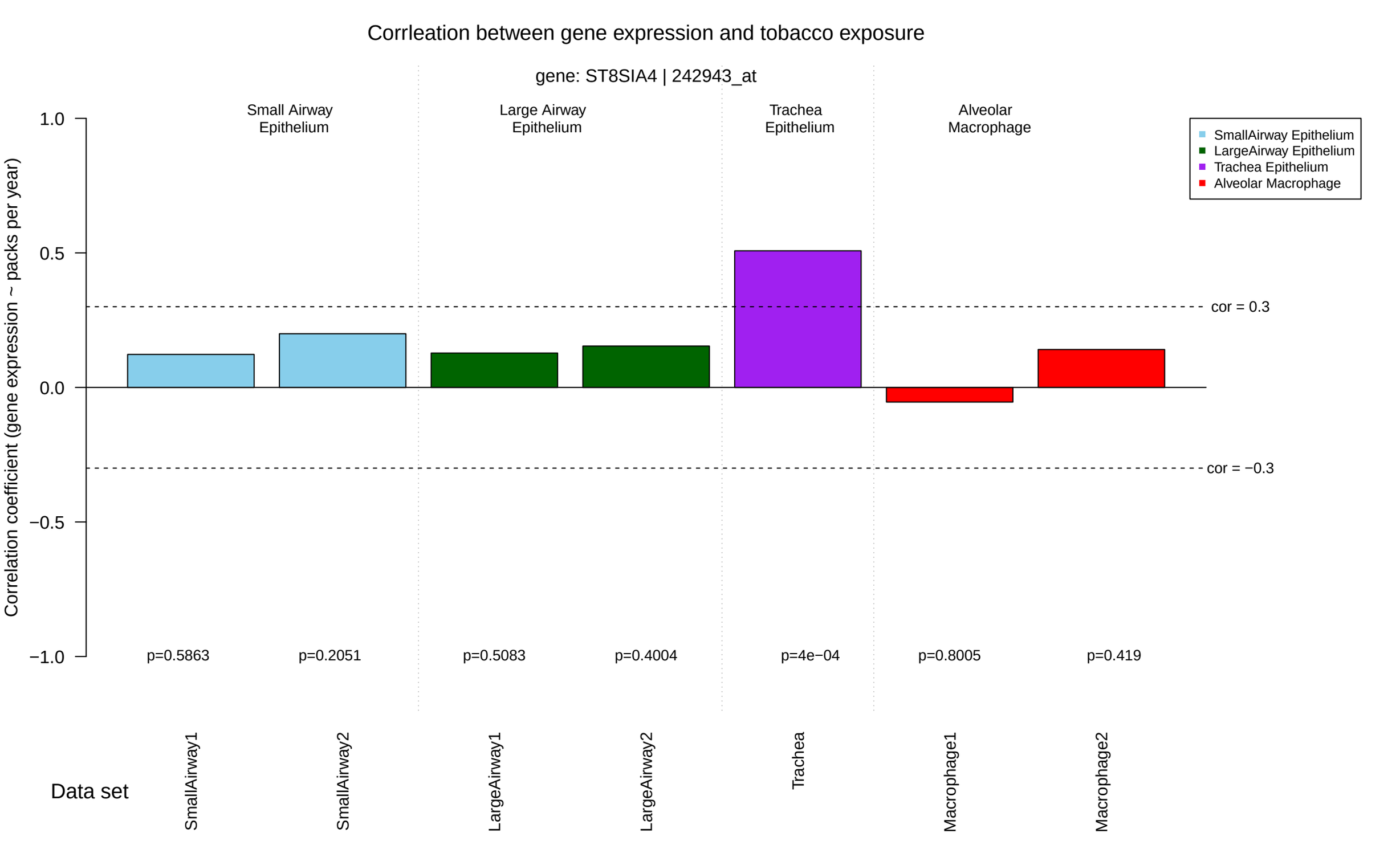

set: LargeAirway1 (p=0.5229)
ST8SIA4 | 242943_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1193)
ST8SIA4 | 242943_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
ST8SIA4 | 242943_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2656)
ST8SIA4 | 242943_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4401)
ST8SIA4 | 242943_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST8SIA4 | 242943_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5863
p=0.2051
p=0.5083
p=0.4004
p=4e−04
p=0.8005
p=0.419
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2